

A moving-endpoint obstruction to all-height second-tail positivity

Analytic existence theorem with two exact finite-sign checks

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Verdict: ADVANCE — a rigorously supported obstruction theorem.

The supplied coefficient-5/4 infinite-height cone survives the present reconstruction. The stronger proposed route, strict or non-strict second-tail positivity at every height and every nonnegative endpoint, does not: the argument in this report proves negative second tails at arbitrarily large heights.

Theorem 1 (Main obstruction). *Use the actual-theta definitions in Section 2. For every $H > 0$, there is a real $t > \max\{H, 448\pi\}$ such that, with*

$$R = \frac{1}{2} \log \frac{t}{448\pi} > 0, \quad C_* = \frac{\pi^3}{2(2\pi)^{47/16}},$$

one has

$$V_{2,t}(R) < -\frac{C_*}{100} t^{63/16} e^{-\pi t/2} < 0.$$

This is an *existence theorem*, not a directly evaluated negative pair (t, R) . No numerical first negative height or effective threshold is asserted. Its proof combines a phase-uniform Mellin scaling limit, a continuously recurring prime-phase pattern, and a finite negative divisor polynomial verified by exact integer/rational arithmetic.

The negative endpoints satisfy $2\pi e^{2R} = t/224$, whereas the predecessor cone starts at $2\pi e^{2R} = 5t/4$. Thus the statements concern different regions. The finite-height [33, 34] theorem is neither contradicted nor newly fully replayed here.

Relationship to RH. The result blocks this particular blanket V_2 -positivity strategy. It proves no off-critical-line zeta zero and does not decide the weaker aggregate-transform sign. It is not an RH proof or disproof.

Review limits. One assistant performed the source-informed analysis and wrote both exact-arithmetic implementations. Four parallel numerical jobs were exploratory only; they were not independent specialist agents. No independent human review or formal proof-kernel verification occurred. Historical novelty is not claimed.

Evidence basis. Incoming cone ZIP SHA-256:

aa03fb1c204d622857f89804a05d5e5f458

0d11bc291410dcff7dde332dc273c

The 72,771,333 bytes, ZIP CRCs, and all 114 listed payload hashes were checked. Hashes identify evidence; they are not mathematical premises.

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1 Scope, starting state, and the reviewed predecessor

The latest requested basis is the supplied RH_001H_V2_Unbounded_Height_Cone_2026-10-04 archive. Its next obligation is to determine the second-tail sign in the moving compact region below the positive cone, or to establish a negative second tail. The present run takes that negative-result branch; it does not certify another adjacent height band.

At the first inventory, this runtime contained the conversation uploads but no preserved local successor workspace from the interrupted request. The earlier agent notes and outputs were available inside the source archive, not as work newly completed in this run. The original cone archive and earlier uploaded archives were left unchanged. An isolated derivative workspace was created.

1.1 What was verified here

The incoming archive has the SHA-256 on the title page, matches the supplied sidecar and integrity JSON, has valid CRCs, and matches all 114 payload manifest entries. The only payload-list exclusion is the manifest itself. The nested immediate affine predecessor retains SHA-256

bb01dc6baacc6ff4d723bdd78a03b525b914e1a2564043e4f8e44eb30202d0.

The cone proof was reconstructed from its integral, phase-loss, ODE, and complete product-remainder arguments. A freshly written exact-rational checker confirms its twelve selected finite comparisons. No mathematical defect was found in that cone proof. Section 8 records the decisive chain rather than promoting a matching executable output to a proof.

1.2 What was not verified here

Python 3.13.5 was available, but python-flint was not installed. Installation failed because the package source was unreachable. The failure log is preserved. Therefore this run did *not* reproduce the Arb endpoint integrals or the full 800-rectangle [33,34] certificate. Those remain inherited claims with their disclosed software trust model. They are not premises of Theorem 1.

The incoming provenance exceptions are retained, not repaired: an earlier D structural output/log was overwritten; a cone A inline diagnostic is a later transcription; and B's first cone check lacked a pre-run hash and stderr. These qualifications concern execution history. The new argument below starts from the explicit definitions and does not rely on those records.

2 Definitions and exact normalization

For real r, t , define

$$\Theta(r) = \sum_{n \geq 1} \left(2\pi^2 n^4 e^{9r/2} - 3\pi n^2 e^{5r/2} \right) e^{-\pi n^2 e^{2r}}, \quad (1)$$

$$\Phi_r(t) = \int_{\mathbb{R}} \Theta(r + u/2) \Theta(r - u/2) e^{-itu} du, \quad (2)$$

$$g_t(r) = 2 \sinh(7r/8) \Phi_r(t), \quad (3)$$

$$V_{2,t}(R) = \int_R^\infty (s - R) g_t(s) ds, \quad R \geq 0. \quad (4)$$

The kernel is real and even in t . Write $d(p)$ for the number of positive divisors of the positive integer p ; p will not denote only primes. Put

$$C_p(t) = \sum_{d|p} \cos\left(t \log \frac{p}{d^2}\right), \quad (5)$$

$$B_t(z) = (z^2 + 9)K_{it}(z) + 6zK'_{it}(z), \quad z > 0. \quad (6)$$

The prime on K means differentiation in its positive argument.

Lemma 2 (Product expansion and convergence). *The definitions above give*

$$g_t(r) = 4\pi^2 e^{5r} \sinh(7r/8) \sum_{p \geq 1} p^2 C_p(t) B_t(2\pi p e^{2r}). \quad (7)$$

For $r \geq 0$ this series converges locally uniformly and absolutely, uniformly in real t . There is a constant independent of t such that, for all sufficiently large r ,

$$|g_t(r)| \leq C \exp\left(\frac{71r}{8} - 2\pi e^{2r}\right). \quad (8)$$

Thus (4) exists and every fixed exponential or polynomial weight needed below is integrable.

Proof. Expand the two theta factors. For their (m, n) term, set $p = mn$, $z = 2\pi e^{2r}$ and $u = v + \log(n/m)$. Direct multiplication of the two quadratic prefactors gives

$$\pi^2 p^2 e^{5r} [(pz)^2 + 9 - 6pz \cosh v] e^{-pz \cosh v}.$$

The Fourier phase additionally contributes $e^{-it \log(n/m)}$. The Bessel integral and its derivative [1] give

$$\int_{\mathbb{R}} e^{-w \cosh v} e^{-itv} dv = 2K_{it}(w), \quad \int_{\mathbb{R}} \cosh v e^{-w \cosh v} e^{-itv} dv = -2K'_{it}(w).$$

Pairing divisors d and p/d makes the phase sum the real number $C_p(t)$. This gives $\Phi_r(t) = 2\pi^2 e^{5r} \sum p^2 C_p B_t(pz)$ and hence (7).

For completeness, $|K_{it}(w)| \leq K_0(w) \leq q(w)$, with $q(w) = e^{-w} \sqrt{\pi/(2w)}$. Also $|K'_{it}(w)| \leq K_1(w) \leq q(w)(1 + 1/(2w))$. The first Gaussian estimate uses $\cosh v \geq 1 + v^2/2$; for the second substitute $y = \sinh(v/2)$ and bound $(1 + 2y^2)/\sqrt{1 + y^2}$ by $1 + 2y^2$. Consequently

$$|B_t(w)| \leq q(w)(w^2 + 6w + 12). \quad (9)$$

The same majorant bounds the absolute expanded theta integrals, so their interchanges were legitimate. Since $|C_p| \leq d(p) \leq p$, the normalized p -term is bounded by a constant times $p^{9/2} e^{-(p-1)z}$ for $z \geq 2\pi$. This is summable. The outer factor grows at most as $e^{47r/8}$, and $z^2 q(z)$ is a constant times $e^{3r-2\pi e^{2r}}$, proving (8). $\square$

3 An exact moving-endpoint integral

Fix $X > 1$. For $t > 0$ sufficiently large set

$$R_X(t) = \frac{1}{2} \log \frac{t}{2\pi X}. \quad (10)$$

We require $t \geq 2\pi X$ so that this endpoint is nonnegative. For the p -term, let $a_p = 2\pi p$ and $z = a_p e^{2s}$. Then $z_0 = a_p e^{2R_X(t)} = pt/X$ and $s - R_X(t) = \frac{1}{2} \log(z/z_0)$. Applying $ds = dz/(2z)$ to (7) gives exactly

$$V_{2,t}(R_X(t)) = \pi^2 \sum_{p \geq 1} p^2 C_p(t) \int_{pt/X}^{\infty} \frac{1}{2} \log \frac{Xz}{pt} B_t(z) \cdot \left[(2\pi p)^{-47/16} z^{31/16} - (2\pi p)^{-33/16} z^{17/16} \right] dz. \quad (11)$$

No integration by parts, discarded upper endpoint, or asymptotic replacement has been made here.

Put

$$\alpha_1 = \frac{31}{16}, \quad \alpha_2 = \frac{17}{16}, \quad \sigma_j = \alpha_j - 1,$$

and, for either pair (α, σ) , define

$$\mathcal{J}_{\alpha,\sigma}(t; X) = \frac{1}{2} \sum_{p \geq 1} p^{-\sigma} C_p(t) \int_{pt/X}^{\infty} z^{\alpha} \log \frac{Xz}{pt} B_t(z) dz. \quad (12)$$

Then (11) is

$$V_{2,t}(R_X(t)) = \pi^2 \left[(2\pi)^{-47/16} \mathcal{J}_{31/16,15/16} - (2\pi)^{-33/16} \mathcal{J}_{17/16,1/16} \right]. \quad (13)$$

The two powers, including the minus sign of the lower-order term, are retained throughout.

4 A phase-uniform Mellin scaling theorem

Define the finite divisor polynomial

$$P_{\sigma}(t; X) = \sum_{1 \leq n < X} n^{-\sigma} C_n(t) \log(X/n). \quad (14)$$

An integer $n = X$ may be included with weight zero when X is an integer.

Theorem 3 (Moving-endpoint scaling). *For every fixed $X > 1$,*

$$\frac{e^{\pi t/2} V_{2,t}(R_X(t))}{C_* t^{63/16}} = P_{15/16}(t; X) + o(1), \quad t \longrightarrow +\infty, \quad (15)$$

where $C_* = \pi^3/[2(2\pi)^{47/16}] > 0$. The error tends to zero with a bound independent of the changing divisor phases. More precisely there are $e_j(t; X) \rightarrow 0$ such that

$$\begin{aligned} \frac{e^{\pi t/2} V_{2,t}(R_X(t))}{C_* t^{63/16}} &= P_{15/16}(t; X) + e_1(t; X) \\ &\quad - \left(\frac{2\pi}{t} \right)^{7/8} \left[P_{1/16}(t; X) + e_2(t; X) \right]. \end{aligned} \quad (16)$$

No assumption that the phases converge is made.

4.1 The complete Mellin moment, including the derivative term

For $\operatorname{Re} \mu > 0$, the classical Mellin integral [3] is

$$M_t(\mu) := \int_0^\infty z^{\mu-1} K_{it}(z) dz = 2^{\mu-2} \Gamma\left(\frac{\mu+it}{2}\right) \Gamma\left(\frac{\mu-it}{2}\right). \quad (17)$$

Its stated condition $|\operatorname{Re}(it)| < \operatorname{Re} \mu$ is satisfied. It can also be derived directly from the Schlaefli integral [2]: substitute that representation for K , exchange the two absolutely convergent integrals, and integrate $z^{\mu+it-1} e^{-z^2/(4u)}$. This produces $2^{\mu-2} \Gamma((\mu+it)/2)$ times the remaining gamma integral $\Gamma((\mu-it)/2)$. All arguments z, u in this derivation are positive.

Integration by parts, retaining the boundary $[z^\mu K_{it}(z)]_0^\infty = 0$, and the gamma recurrence now give

$$\begin{aligned} \int_0^\infty z^{\mu-1} B_t(z) dz &= M_t(\mu+2) + (9-6\mu)M_t(\mu) \\ &= [t^2 + (\mu-3)^2] M_t(\mu). \end{aligned} \quad (18)$$

The boundary at zero vanishes because $|K_{it}(z)| \leq K_0(z) = O(1 + |\log z|)$; the boundary at infinity vanishes exponentially. Alternatively these small- z bounds follow immediately from the positive K_0 integral by splitting at $\log(2/z)$. The derivative term is integrable since $|K'_{it}| \leq K_1$.

4.2 Mellin inversion without a finite product truncation

The elementary inversion identity, for $y > 0$ and $d > 0$, is

$$(\log y)_+ = \frac{1}{2\pi i} \int_{d-i\infty}^{d+i\infty} \frac{y^s}{s^2} ds. \quad (19)$$

This integral is absolutely convergent. Close a semicircle to the left when $y > 1$, obtaining the residue $\log y$ at zero; close to the right when $y < 1$, obtaining zero. The semicircle contribution is $O(1/L)$ because the exponential is bounded on the appropriate semicircle. At $y = 1$ the integral of $1/s^2$ is zero by its antiderivative.

For $\operatorname{Re} w > 1$, let

$$F_t(w) = \sum_{p \geq 1} C_p(t) p^{-w}.$$

The series is absolutely convergent and satisfies the *uniform* bound

$$|F_t(w)| \leq \sum_{p \geq 1} d(p) p^{-\operatorname{Re} w} = \left(\sum_{n \geq 1} n^{-\operatorname{Re} w} \right)^2. \quad (20)$$

Only a convergent positive series is used; no analytic continuation of zeta and no assertion about its zeros is required.

Use (19) with $d = 2$ in (12). Here $\sigma + 2 > 1$. The exchanges of p, z , and the vertical integral are absolute: they are bounded, for each fixed t , by a constant times

$$\sum_{p \geq 1} d(p) p^{-\sigma-2} \int_0^\infty z^{\alpha+2} |B_t(z)| dz \int_{\mathbb{R}} \frac{dv}{4+v^2} < \infty.$$

Therefore, using (18),

$$\mathcal{J}_{\alpha,\sigma} = \frac{1}{2} \frac{1}{2\pi i} \int_{\operatorname{Re} s=2} \frac{X^s F_t(\sigma+s)t^{-s}}{s^2} \cdot [t^2 + (\alpha+s-2)^2] M_t(\alpha+s+1) ds. \quad (21)$$

Define

$$\mathcal{R}_{\alpha,t}(s) = \left(1 + \frac{(\alpha+s-2)^2}{t^2}\right) \frac{M_t(\alpha+s+1)}{\pi e^{-\pi t/2} t^{\alpha+s}}. \quad (22)$$

The positive real logarithm defines $t^{\alpha+s}$. We have the exact identity

$$\frac{\mathcal{J}_{\alpha,\sigma}}{(\pi/2)e^{-\pi t/2} t^{\alpha+2}} = \frac{1}{2\pi i} \int_{\operatorname{Re} s=2} \frac{X^s F_t(\sigma+s)}{s^2} \mathcal{R}_{\alpha,t}(s) ds. \quad (23)$$

4.3 The uniform bound that permits changing phases

Set $s = 2 + iv$ and $u = \alpha + 3 > 1$. Stirling's formula on vertical lines implies, for fixed $u > 1$, a finite constant A_u such that for every real y ,

$$|\Gamma(u/2 + iy)| \leq A_u(1 + |y|)^{(u-1)/2} e^{-\pi|y|/2}. \quad (24)$$

Indeed Stirling's expansion [4] gives the estimate as $|y| \rightarrow \infty$ in sectors strictly inside $|\arg z| < \pi$; both vertical directions are admissible. Continuity on the remaining compact segment supplies a single finite constant. There are no gamma poles there because $u/2 > 0$.

From (17) and the exact identity $|v+t| + |v-t| = 2 \max\{t, |v|\}$, it follows for $t \geq 1$ that

$$\left| \frac{M_t(u+iv)}{\pi e^{-\pi t/2} t^{u+iv-1}} \right| \leq A'_u \left(2 + \frac{|v|}{t}\right)^{u-1} \exp\left[-\frac{\pi}{2}(|v|-t)_+\right]. \quad (25)$$

The constant absorbs fixed powers of two. If $|v| \leq t$, the right side is uniformly bounded. If $|v| > t$, write $w = |v| - t \geq 0$; then $2 + |v|/t \leq 3 + w$. Any fixed power of $3 + w$ times $e^{-\pi w/2}$ is bounded. The other factor in (22) has modulus at most $1 + (\alpha^2 + v^2)/t^2$, which adds only a quadratic factor to the same argument. Hence

$$\sup_{t \geq 1, v \in \mathbb{R}} |\mathcal{R}_{\alpha,t}(2+iv)| < \infty. \quad (26)$$

This treats the regions $v \approx \pm t$ as well as bounded v ; they are not omitted.

For each fixed complex μ with positive real part, Stirling and the fixed-shift gamma ratio give

$$M_t(\mu) \sim \pi e^{-\pi t/2} t^{\mu-1}, \quad t \rightarrow +\infty. \quad (27)$$

For example first use conjugate gamma factors for real $\mu = u$, then apply the gamma ratio [5] to the fixed shifts $iv/2$ of arguments $u/2 \pm it/2$. The two principal-power phase factors cancel their real exponentials and leave $(t/2)^{iv}$; the outside 2^{iv} leaves t^{iv} . Thus there is no missing phase or sign in (27). It follows that $\mathcal{R}_{\alpha,t}(2+iv) \rightarrow 1$ for every fixed v .

Combining (20) with (26), the error on replacing $\mathcal{R}$ by one in (23) is at most

$$\frac{X^2}{2\pi} \left(\sum_{n \geq 1} n^{-\sigma-2} \right)^2 \int_{\mathbb{R}} \frac{|\mathcal{R}_{\alpha,t}(2+iv) - 1|}{4 + v^2} dv \rightarrow 0. \quad (28)$$

This is dominated convergence applied to an integrable majorant independent of t . Crucially, it does not require F_t or its phases to converge. It is an actual uniform error bound in those phases, not a subsequence assumption.

With $\mathcal{R}$ replaced by one, (19), termwise integration, and (20) give exactly $P_\sigma(t; X)$. Thus

$$\mathcal{J}_{\alpha,\sigma} = \frac{\pi}{2} e^{-\pi t/2} t^{\alpha+2} [P_\sigma(t; X) + e_\alpha(t; X)], \quad e_\alpha \rightarrow 0.$$

Substitution in (13) proves (16). For fixed X , $P_{1/16}$ is bounded by a finite sum using $|C_n| \leq d(n)$, so its coefficient $(2\pi/t)^{7/8}$ tends to zero. This proves Theorem 3.

4.4 Separate sign and endpoint check

The derivative term can be checked before Mellin inversion. For $w(z) = \frac{1}{2} \log(z/z_0)$, direct integration by parts gives

$$\int_{z_0}^{\infty} z^\alpha w(z) B_t(z) dz = \int_{z_0}^{\infty} \left[z^{\alpha+2} w(z) + (3 - 6\alpha) z^\alpha w(z) - 3z^\alpha \right] K_{it}(z) dz. \quad (29)$$

The lower boundary is zero because $w(z_0) = 0$; the upper one vanishes exponentially. Applying Mellin inversion to the log-weighted terms and the step weight to the last term produces $M_t(\mu + 2) + (3 - 6\alpha - 6s)M_t(\mu)$, where $\mu = \alpha + s + 1$. Since $3 - 6\alpha - 6s = 9 - 6\mu$, this agrees with (18). This separately checks the sign and the factor $1/2$ in the log weight. It is an algebraic cross-check, not a second independent proof of the uniform-limit lemma.

5 An exact negative finite polynomial

Let $\Omega(n)$ count prime factors with multiplicity and put $\lambda(n) = (-1)^{\Omega(n)}$. At $X = 224$ define

$$D = \sum_{n=1}^{223} d(n) \lambda(n) n^{-15/16} \log(224/n), \quad (30)$$

$$B = \frac{1}{2} \sum_{n=1}^{223} d(n) \Omega(n)^2 n^{-15/16} \log(224/n). \quad (31)$$

Two separate implementations prove the strict comparisons

$$D < -\frac{3}{100}, \quad B < 156. \quad (32)$$

These are the only machine-checked numerical inequalities needed for the new obstruction. They concern 223 elementary terms, not values of V_2 .

5.1 Certificate A: integer roots and logarithm series

Use $Q = 2^{100}$. For each n , integer bisection finds a_n with

$$a_n^{16} \leq nQ^{16} < (a_n + 1)^{16}.$$

This encloses $n^{1/16}$, and $n^{-15/16} = n^{1/16}/n$. Exact sixteenth powers verify every root witness.

Write $224/n = 2^k y$ with $1 \leq y < 2$. For $u = (y - 1)/(y + 1) \in [0, 1/3]$ and $N = 36$,

$$0 \leq \log y - 2 \sum_{j=0}^{N-1} \frac{u^{2j+1}}{2j+1} \leq \frac{2u^{2N+1}}{(2N+1)(1-u^2)}. \quad (33)$$

This follows by integrating the geometric series for $1/(1-u^2)$, with a positive tail. The same bound at $y = 2$ encloses $\log 2$. All arithmetic before outward fixed-point rounding is rational. Signed summation uses the correct lower/upper swap for every negative term.

The resulting rational enclosures, displayed with outward-rounded decimal endpoints, are

$$\begin{aligned} D &\in [-0.032763882323341271960959, -0.032763882323341271960958], \\ B &\in [155.705129275089390978560725, 155.705129275089390978560726]. \end{aligned}$$

The exact governing integers for D are

$$\begin{aligned} Q &= 1267650600228229401496703205376, \\ QD_{\text{low}} &= -41533155092990638659436879387, \\ QD_{\text{up}} &= -41533155092990638659436879099. \end{aligned}$$

Here D_{low} and D_{up} denote the enclosure endpoints. Complete root, log, factorization, and weight witnesses are in `results/divisor_certificate.json`.

5.2 Certificate B: a different exact logarithm method

The second implementation imports no code from the first. It counts divisors directly, counts Ω by a prime-power sieve, and verifies the root witnesses by exact powers. It does not reuse the first logarithm calculation. Instead it integrates $1/u$ from 1 to y by exact rational composite Simpson quadrature with 64 panels. For $h = (y - 1)/64$,

$$|\log y - S_{64}(y)| \leq \frac{(y-1)h^4}{180} \sup_{[1,y]} \left| \frac{24}{u^5} \right| \leq \frac{2}{15}(y-1)h^4. \quad (34)$$

This is the composite Simpson remainder in DLMF 3.5.8 [7], with an even panel count and $f(u) = 1/u \in C^4[1, y]$. The fourth-derivative bound is explicit and the interval is contained in $[1, 2]$.

With outward 2^{-60} rounding, this produces coarser but independent logarithm enclosures. Conservative decimal displays are

$$-0.032764412 < D < -0.032763353, \quad 155.7051281 < B < 155.7051311.$$

The exact scaled integer output, rather than these short displays, governs the strict tests. Both (32) pass. The second enclosure also contains the first enclosure.

Independence qualification. The two implementations use different logarithm algorithms and independently reconstruct divisor counts and multiplicities. They share Python's integer/rational arithmetic, the mathematical definitions, and the first file's root witnesses (which are fully rechecked). They were written by the same assistant. They are not independent human or agent reviews.

6 Prime phases transfer the finite sign

Lemma 4 (A robust negative phase neighborhood). *If, for every prime $q < 224$,*

$$\text{dist}(t \log q, \pi + 2\pi\mathbb{Z}) \leq \frac{1}{100}, \quad (35)$$

then

$$P_{15/16}(t; 224) < -\frac{9}{625}. \quad (36)$$

Proof. Write $t \log q = \pi + 2\pi k_q + \varepsilon_q$ with $|\varepsilon_q| \leq 1/100$. For $n < 224$ and $d \mid n$,

$$t \log(n/d^2) \equiv \pi(\Omega(n) - 2\Omega(d)) + \delta_{n,d} \pmod{2\pi},$$

where

$$|\delta_{n,d}| \leq \sum_{q \mid n} |\nu_q(n) - 2\nu_q(d)| |\varepsilon_q| \leq \frac{\Omega(n)}{100}.$$

Thus each divisor cosine is $\lambda(n) \cos \delta_{n,d}$. The inequality $|1 - \cos u| \leq u^2/2$ gives

$$|C_n(t) - d(n)\lambda(n)| \leq \frac{d(n)\Omega(n)^2}{2 \cdot 10000}.$$

All weights in (14) are nonnegative, so

$$|P_{15/16}(t; 224) - D| \leq \frac{B}{10000}.$$

Using (32),

$$P_{15/16}(t; 224) < -\frac{3}{100} + \frac{156}{10000} = -\frac{9}{625}.$$

□

Lemma 5 (Arbitrarily large recurring phase choices). *The open version of (35) is satisfied at arbitrarily large positive real t .*

Proof. There are 48 primes below 224. Their logarithms are rationally independent: if $\sum m_q \log q = 0$ for integers m_q , exponentiation and unique prime factorization force all $m_q = 0$.

Here is the continuous-time equidistribution argument needed, rather than an assumption of independently chosen phases. For every nonzero integer vector m the frequency $\omega_m = \sum m_q \log q$ is nonzero, and

$$\frac{1}{T} \int_0^T e^{it\omega_m} dt = \frac{e^{iT\omega_m} - 1}{iT\omega_m} \longrightarrow 0.$$

Choose a nonnegative C^2 periodic bump supported inside the arc of radius $1/100$ around π , with positive integral. Its Fourier coefficients are absolutely summable, by two integrations by parts; its Fourier series represents the bump by Fourier convergence (or uniqueness) [8]. The product of 48 such bumps has an absolutely summable Fourier series. Averaging that series along $t \mapsto (t \log q)_q$ term by term leaves only its strictly positive constant coefficient in the limit. Thus the flow enters the open phase box at arbitrarily large positive times. If all entries occurred below some fixed time, that positive time-average limit would instead be zero.

This proof concerns real continuous time. It does *not* invoke a discrete integer-time density assertion, which would require a different independence condition involving 2π . □

7 Completion of the obstruction proof

By Theorem 3, there is a finite T_0 such that for every $t > T_0$,

$$\left| \frac{e^{\pi t/2} V_{2,t}(R_{224}(t))}{C_* t^{63/16}} - P_{15/16}(t; 224) \right| < \frac{1}{250}.$$

The existence of this single threshold follows from the phase-uniform bound (28); it is not a different threshold chosen for each phase.

Given $H > 0$, Lemma 5 supplies a $t > \max\{H, 448\pi, T_0\}$ satisfying the phase box. Lemma 4 then gives

$$\frac{e^{\pi t/2} V_{2,t}(R_{224}(t))}{C_* t^{63/16}} < -\frac{9}{625} + \frac{1}{250} = -\frac{13}{1250} < -\frac{1}{100}.$$

Since the normalizing factor is strictly positive and $R_{224}(t) > 0$, this proves Theorem 1. Evenness yields a reflected statement at negative heights.

Scope. No numerical value for T_0 , a recurrence time, or a first negative height is computed. This is a complete existential implication, not a direct finite-height special-function certificate. An effective or first-counterexample localization remains a separate task. The number 224 is a successful exact witness, not a proved minimal cutoff.

8 Review of the incoming cone and compatibility

The incoming theorem states, for $\tau = |t| \geq 34$,

$$z = 2\pi e^{2r} \geq \frac{5}{4}\tau \implies g_t(r) > 0, \quad R \geq \frac{1}{2} \log \frac{5\tau}{8\pi} \implies V_{2,t}(R) > 0.$$

The following reconstruction checks the infinite-parameter mechanism.

Set $\theta = \arcsin(\tau/z)$ and $a = \sqrt{z^2 - \tau^2} \geq 3\tau/4 \geq 51/2$. Shifting the full-line Bessel integral to height $i\theta$ gives the exact real expression

$$K_{i\tau}(z) = e^{-\tau\theta} \int_0^\infty e^{-a \cosh u} \cos\{\tau(u - \sinh u)\} du.$$

The vertical sides are bounded by $\theta e^{-a \cosh L}$ and vanish. The full adverse phase contribution is at most

$$\frac{5\tau^2 e^{-a} \sqrt{\pi/2}}{24(a-2)^{7/2}}.$$

This follows from $0 \leq \sinh u - u \leq u^3 \cosh u/6$, $1 - \cos v \leq v^2/2$, and the full sixth Gaussian moment; it is not a truncation of an oscillatory integral. A nonoscillatory lower bound is $K_0(a) \geq e^{-a-1}/\sqrt{a}$. The source's elementary rational comparisons give

$$e^a \sqrt{a} L < \frac{393040}{14639043} < \frac{1}{36}, \quad e < \frac{11}{4}, \quad \pi < \frac{32}{9},$$

hence

$$K_{i\tau}(z) > \frac{133}{396} \frac{e^{-a-\tau\theta}}{\sqrt{a}} > \frac{1}{3} \frac{e^{-a-\tau\theta}}{\sqrt{a}}.$$

Using $\theta < 1$, $\tau \leq 4z/5$, and $a \leq z$ yields $K_{i\tau}(z)/q(z) > \frac{1}{4} e^{-4z/5}$.

For fixed τ put $y(w) = \sqrt{w}K_{i\tau}(w)$. The Bessel ODE [6] is $y'' = (1 - (\tau^2 + 1/4)/w^2)y$. Positivity has already been proved on the relevant cone. The quantity $W = e^{-w}(y' + y)$ has negative derivative and zero limit at infinity, so $W > 0$. This gives $K'/K > -1 - 1/(2w)$ and

$$B_t(w) > (w^2 - 6w + 6)K_{it}(w).$$

The estimates in Lemma 2, including all $p \geq 2$ terms with $|C_p| \leq d(p) \leq p$, give the full normalized remainder

$$E(z) = \left(1 + \frac{6}{z} + \frac{12}{z^2}\right) \frac{2^{9/2}e^{-z}}{1 - (3/2)^{9/2}e^{-z}} < 96e^{-z}, \quad z \geq 85/2.$$

The leading normalized block exceeds $\frac{1}{8}e^{-4z/5}$. Its strict dominance follows from

$$e^{z/5} \geq e^{17/2} > \sum_{k=0}^6 \frac{(17/2)^k}{k!} = \frac{58017253}{46080} > 768.$$

All omitted products are charged, every denominator is positive, and $z > 2\pi$ makes the sinh prefactor positive. The tail integral is strictly positive including at the cutoff, by (8). This reconstruction finds no defect in the cone theorem.

For the new negative endpoints, however,

$$2\pi e^{2R_{224}(t)} = \frac{t}{224} < \frac{5}{4}t.$$

Their radial separation from the cone cutoff is exactly $\frac{1}{2} \log 280$. Thus no seam failure or contradiction with the positive cone occurs. The obstruction is inside the previously uncontrolled region. No existing finite-height compact certificate is used or invalidated.

9 RH relationship and the remaining viable obligation

The source record supplies the theta-to-xi identity at the fixed secant center 15/16. Retaining its normalization, it reads

$$\begin{aligned} \hat{H}_x(t) &= \int_0^\infty \sinh(2xr)g_t(r) \, dr \\ &= \frac{1}{8} \left(|\xi(15/16 + x + it)|^2 - |\xi(15/16 - x + it)|^2 \right). \end{aligned} \quad (37)$$

That theta-to-xi identification remains an explicitly inherited identity in this report; it is not needed to prove the obstruction. The standard definition and functional equation of ξ are in [9]. The second-tail reformulation itself follows directly here: $V_2'' = g$, and two integrations by parts, with upper boundaries killed by (8), give

$$\hat{H}_x(t) = 2xV_{2,t}(0) + 4x^2 \int_0^\infty \sinh(2xr)V_{2,t}(r) \, dr. \quad (38)$$

At the lower boundary the first term is $2xV_2(0)$, not zero. This holds for fixed real x , in particular $0 < x \leq 1/16$.

A negative V_2 at some positive endpoint does not determine the sign of (38). Positive and negative contributions can coexist, and the boundary term remains separate. Thus Theorem 1 does not prove a negative secant transform and does not locate any zero of ξ .

The blanket sufficient hypothesis $V_{2,t}(R) \geq 0$ for every $t \in \mathbb{R}$ and $R \geq 0$ is now obstructed. Merely improving affine seeds or backward-transport error budgets cannot establish that false global statement. The unresolved weaker target is a direct proof, or falsification, of

$$\hat{H}_x(t) > 0 \quad \text{for all real } t \text{ and } 0 < x \leq 1/16,$$

with the signed compact contribution and boundary term included rather than replaced by pointwise V_2 positivity. This report does not settle it.

Even if that aggregate statement were proved, (37) would exclude a zero only at right secant points with $15/16 < \operatorname{Re} s \leq 1$. Reflection would confine zeros to the corresponding initial strip, with boundary issues separate. No valid unconditional descent from that fixed strip to the critical line is supplied here. The obstruction clarifies a route that cannot work; it is not a completed RH implication.

10 Adversarial audit, trust, and replay

10.1 Attacks resolved in the written argument

Attack	Resolution and exact scope
Moving phases inside a limit	Equation (28) controls the error uniformly in F_t ; no phase convergence is assumed.
Uncontrolled vertical frequencies	Equation (25) covers $v \approx \pm t$ and $ v > t$, not just bounded v .
Uncharged infinite product tail	The full absolutely convergent $F_t(\sigma + s)$ stays inside the Mellin integral; the finite polynomial appears by exact inversion, not truncation.
Dropped derivative or endpoint	Equations (18) and (29) agree; lower log weight and both limiting endpoints are explicit.
Wrong exponent or normalization	The two exact terms in (11) give $63/16$, $15/16$, and the lower-order factor $(2\pi/t)^{7/8}$.
Finite signs substituted for recurrence	Lemma 5 proves arbitrarily large continuous-time visits to the entire 48-prime neighborhood.
Discrete/continuous quantifier confusion	The Fourier average is an integral over real time, not an integer-time sequence.
Only nonnegativity checked	The finite bound is $P < -9/625$; the final error budget yields $V_2 < -C_* t^{63/16} e^{-\pi t/2} / 100$.
Contradiction with positive cone	The negative ray is $z = t/224$; the positive cone is $z \geq 5t/4$.
Hash treated as proof	Integrity, exact finite computations, and universal analytic lemmas have separate records and trust scopes.

10.2 Preserved diagnostics and limitations

Four parallel mpmath/Simpson reconnaissance jobs at $t = 35, 40, 50, 80$ found no negative sampled second tail on their selected 240-point grids. Their product/radial truncations and quadrature were not rigorously bounded. They are preserved as exploratory results only, not premises or certifications. A subsequent exploratory Liouville-divisor calculation suggested the successful $X = 224$ witness; both exact certificates were then constructed without relying on the exploratory error estimates.

The analytic proof trusts ordinary real/complex analysis, the verified primary Bessel and gamma formulas, elementary Fourier arguments, and the displayed finite inequalities. The finite inequalities trust Python’s integer and Fraction arithmetic and the correctness of the reviewed enclosure code. No floating-point arithmetic or special-function library enters those comparisons. The second checker uses floating point only for explicitly labeled display fields after the exact comparisons; those fields are not proof premises.

The universal analytic statements have not been verified in a formal proof assistant. The same assistant produced and adversarially examined them; no independent external mathematician or second agent has approved the proof. A first negative height, effective recurrence bound, effective asymptotic threshold, historical priority, and a new aggregate-transform theorem remain unresolved.

10.3 Portable replay and evidence map

From the extracted successor root, ordinary Python 3 is sufficient for the finite checks; do not use `-O`, because assertions are part of the verification. Run

```
python code/replay.py /path/to/a/new/replay-directory
```

The runner refuses an existing destination, regenerates the tight certificate, compares mathematical records excluding the timestamp, runs the separate Simpson checker, and checks the cone’s finite constants. It does not attempt an Arb replay and does not claim to machine-prove the analytic theorem.

Claim	Material
Negative finite D and sensitivity	<code>code/certify_divisor_polynomial.py</code> , <code>results/divisor_certificate.json</code> .
Separate finite reconstruction	<code>code/check_certificate_independent.py</code> , <code>results/independent_certificate_check.json</code> .
Universal obstruction	This report, Sections 2–5 and the phase/recurrence/completion sections.
Cone review	Section 8, retained source proof, <code>results/cone_constant_review.json</code> .
Starting evidence and limits	<code>provenance/</code> , <code>review/REVIEW_RECORD.md</code> , <code>DEPENDENCY_MAP.json</code> .
Exploratory negatives/positives	<code>diagnostics/</code> ; all explicitly non-certifying.
Byte identity	Root <code>MANIFEST.sha256.json</code> ; external ZIP sidecar and integrity JSON.

Final mathematical classification. The finite-height predecessor is not disproved; its full new independent numerical audit is not claimed. The incoming analytic cone is supported by the current reconstruction. The proposed global second-tail sign is obstructed by a new analytic existence theorem with an exact finite witness. No public repository was modified and nothing was submitted or sent externally.

References

- [1] NIST Digital Library of Mathematical Functions, Eq. 10.32.9, modified-Bessel integral. <https://dlmf.nist.gov/10.32.E9>. Used for positive arguments and imaginary order.
- [2] NIST DLMF, Eq. 10.32.10, Schlaefli integral. <https://dlmf.nist.gov/10.32.E10>. Its $|\arg z| < \pi/4$ condition holds since $z > 0$.
- [3] NIST DLMF, Eq. 10.43.19, Mellin transform of K_ν . <https://dlmf.nist.gov/10.43.E19>. Used with $\nu = it$ and $\operatorname{Re} \mu > 0$.
- [4] NIST DLMF, §5.11(i)–(ii), especially Eqs. 5.11.3 and 5.11.10, Stirling expansion and remainder. <https://dlmf.nist.gov/5.11>. Applied on vertical lines with fixed positive real part, away from the negative-real sector boundary.
- [5] NIST DLMF, Eq. 5.11.12, gamma ratio with fixed complex shifts in $|\arg z| \leq \pi - \delta$. <https://dlmf.nist.gov/5.11.E12>. Both vertical rays used here satisfy this condition.
- [6] NIST DLMF, Eq. 10.25.1, modified-Bessel differential equation. <https://dlmf.nist.gov/10.25.E1>.
- [7] NIST DLMF, Eqs. 3.5.7–3.5.8, composite Simpson rule and its fourth-derivative remainder. <https://dlmf.nist.gov/3.5.E8>. Applied to $1/u$ on a positive compact interval, with 64 panels.
- [8] NIST DLMF, §1.8(i)–(ii), Fourier coefficient decay, uniqueness, and pointwise convergence. <https://dlmf.nist.gov/1.8>. The periodic bump used here is C^2 , so it satisfies the convergence hypotheses and has absolutely summable coefficients.
- [9] NIST DLMF, Eqs. 25.4.3–25.4.4, completed zeta function and functional equation. <https://dlmf.nist.gov/25.4>. This is contextual, not a premise of the obstruction.